bladder management, extending the skills of the veterinary therapist and dictating close interaction between the therapist and veterinarian.

Therapeutic modalities and exercises prescribed depend on the stage of the postoperative period and the neurologic status of the patient. Control of pain and swelling and the beginning of exercise are the focus of the initial postoperative period. Swelling of the surgical site is addressed with cryotherapy. Electrical stimulation is applied for pain relief using interferential or premodulated interferential waveforms. Therapeutic ultrasound must be used with care over the thoracic and lumbar epaxial musculature to avoid generating excess heat over the laminectomy site and possibly damaging nerves and the spinal cord.

Depending on the neurologic deficits, the patient may be unable to sit, stand, ambulate, or urinate independently. Appropriate supportive care, including bladder management, must be carefully performed, and there must be close communication between the therapist and veterinarian. Occasionally patients have significant hyperesthesia along the thoracolumbar areas. In these cases, light massage and desensitization procedures may be useful. Patients with significant paraparesis or paraplegia may benefit from therapeutic exercises designed to help gain trunk control. One exercise is to assist patients to a sitting position with decreasing amounts of support until they are able to sit independently. After this is achieved balance can be challenged using gentle perturbation exercises on firm surfaces initially, and then progressing to unstable surfaces such as an exercise roll, foam mats, bubble wrap, or pillows. It may be helpful for the therapist to be positioned caudal to the patient and place the hands on the cranial stifle to help keep them extended to maintain a weight bearing position. Russian or biphasic ES to the muscles of the pelvic limbs may be used to induce stifle and hock extension and flexion, as tolerated by the patient.

Standing exercises may begin as soon as treatment is initiated. Appropriate support is provided while placing the patient in a standing posture to ensure some loading of the pelvic limbs and correct positioning of the feet. To promote correct stance, support the patient under the chest and pelvis and place the limbs in the correct stance twice daily while the patient is eating. This fosters continued extension of the limbs by proprioceptive neuromuscular facilitation and allows the patient to remain in the standing position for a longer period of time. This can be performed on a firm surface or with the patient supported by an exercise roll. Gently allow the dog to intermittently bear weight. If the patient tends to sink down, tickle the perineal region to stimulate extension of pelvic limbs. After the patient is able to maintain a standing position, pressure is applied to the pelvic limbs by pressing on the dorsal aspect of the pelvis. Gentle perturbations are applied from all angles while the patient is standing so that it is challenged to maintain stability and a standing position. Early ambulation may be assisted by moving the pelvic limbs in a reciprocal walking pattern.

Ambulation is allowed at controlled paces in patients with motor control. Assisted tail or sling walking and underwater treadmill hydrotherapy may be used to unload weight before attaining complete ambulatory independence. Strengthening exercises are initiated as independent ambulation improves (Figures 34-13 through 34-17). Activities include ascending and descending inclines, walking in circles or figure eights, walking on terrain with varying textures (carpet, grass, sand), stepping over objects of varying size to regain proprioceptive awareness, and practicing sit-to-stand activities for coordination. Nonslip flooring is ideal to encourage proprioception and appropriate limb placement. Commercially available booties may provide protection for dogs that tend to knuckle over on the dorsum of the paw or drag the paws while ambulating.

After the incision has healed, deep water hydrotherapy may be initiated. Before active movement of the pelvic

Figure 34-13 Facilitation of weight bearing on rear limbs using a therapy roll.

Figure 34-14 Platform used to facilitate limb placement.

Figure 34-15 Platform used to facilitate weight bearing and limb strengthening.

Figure 34-16 An individual limb exercise.

limbs, swimming benefits the trunk muscles, aiding the patient's overall stability. Active movement of the pelvic limbs may be elicited by applying resistance to the paw, stimulating the medial aspect of the stifle, or flexing the hock.

In patients that are unable to urinate voluntarily, preventing urine retention is essential. Initially, this may require intermittent or continuous urethral catheterization or bladder expression. After the catheter is removed the dog is taken outside three times daily to see if the patient can urinate voluntarily. Taking several dogs together will sometimes induce urination as the patient sees and smells other dogs urinate. If the patient does not urinate completely after an adequate amount of time, palpate to check bladder size. If full the bladder is manually expressed. Urine samples are periodically obtained to identify and treat any infection.

Figure 34-17 Dancing exercise.

A minimum of 3 weeks of physical rehabilitation is recommended. Frequently this is extended to continue facilitating recovery. The degree of success in rehabilitation of thoracolumbar disk disease varies greatly and may take several months of dedicated treatment by the therapist and owner.

Degenerative Lumbosacral Stenosis

Degenerative lumbosacral stenosis is a degenerative disorder resulting in stenosis of the vertebral canal causing compression of the lumbosacral nerve roots or cauda equina. In most adult dogs the caudal end of the spinal cord is located at the level of the L6-7 disk. The nerve roots originating in the L6, L7, sacral, and caudal spinal cord segments course from the distal end of the spinal cord to exit their respective foramina. This collection of nerve roots is called the cauda equina.

Degenerative lumbosacral stenosis occurs when soft tissue and bony changes, possibly in conjunction with abnormal motion of the lumbosacral joint, impinge on the nerve roots of the cauda equina. The most common change is protrusion of the L7-S1 disk, but osteophytes of the articular facets and hypertrophy of the interarcuate ligament and articular facet joint capsule may also cause nerve root compression.

Degenerative lumbosacral stenosis is most common in middle-aged, large-breed dogs. German shepherd dogs are the most commonly affected breed. Working dogs and other large-breed dogs, including boxers, rottweilers, and Doberman Pinschers, are also at increased risk.[12]

Pain and lameness are usually more prominent than frank neurologic deficits because stenosis affects the cauda equina, which is more resistant to damage from compressive forces compared with the spinal cord. The most common sign is lumbosacral pain, characterized by reluctance to work or participate in normal activity, run, jump, or climb stairs. Affected dogs often maintain a characteristic posture, keeping their lumbosacral joint flexed, which decreases nerve root compression. Tail weakness, urinary or fecal incontinence, and pelvic limb lameness or weakness can also occur. On examination, palpation or extension of the lumbosacral joint and elevation of the tail usually induces pain. Other findings may include a weak perineal reflex, proprioceptive positioning deficits in one or both pelvic limbs, and decreased hock flexion when eliciting a flexor withdrawal reflex. The differential diagnosis includes other lesions of the lumbosacral region (discospondylitis, neoplasia), degenerative myelopathy, fibrocartilaginous embolic myelopathy, orthopedic diseases (hip arthritis, cruciate rupture), trauma, and urogenital diseases, such as prostatic disease.

Diagnosis is based on clinical features and imaging. Survey radiographs may be normal or reveal chronic nonspecific changes to the lumbosacral region, including sclerosis of the vertebral end plates, ventral spondylosis deformans, and arthritis of the articular facets. Myelography is useful in assessing the caudal region of the lumbar spine, but will not identify lumbosacral stenosis when the dural sac ends cranial to the lumbosacral junction, which happens in about 25% of large-breed dogs. Although contrast radiography may provide evidence of static and dynamic compression, definitive diagnosis and localization are best made on CT and MRI[13] (Figure 34-18).

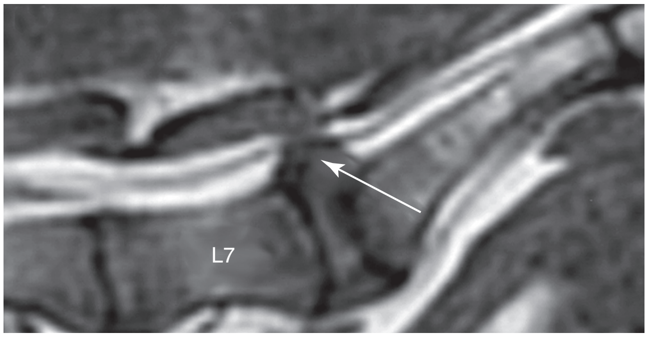

Figure 34-18 Sagittal plane MRI showing protrusion of the L7-S1 disk (*arrow*) with compression of the cauda equine. *L7*, The L7 vertebral body.

Nonsurgical treatment is appropriate initially for patients with mild pain and entails strict rest, antiinflammatory or analgesic medications, and weight loss if necessary. Epidural injections of glucocorticoids improve signs in many patients and avoid the complications associated with long-term administration of oral steroids.[14] Physical rehabilitation also may be useful in the medical management of lumbosacral disease.

Surgery is indicated when there is persistent pain or neurologic deficits. The most common procedure is dorsal laminectomy of L7-S1 with excision of the protruded disk and foraminotomy if necessary. Some surgeons also include fusion of the L7-S1, but it is unclear in which patients this is necessary. Resolution of pain is good, with 80% to 95% of patients improving postoperatively.[15-18] In working dogs, 67% to 80% of patients are able to return to work.[19,20] Only about 25% of patients with incontinence have a return of continence, and more than a 2-week duration of incontinence carries a poor prognosis.[16] Recurrence of clinical signs occurs in about 18% of patients.[15]

Physical Rehabilitation for Degenerative Lumbosacral Stenosis

Physical rehabilitation is beneficial as part of the medical and postoperative management of lumbosacral disease. In combination with medication, therapeutic modalities focusing on pain management, such as ES protocols and therapeutic US, massage therapy, and aquatic therapy are prescribed. Surgical patients are provided routine care of the incision site with icing and limited ES for pain management while hospitalized. Therapists should be informed of the type and location of any implants. Continued care consists of cage or crate confinement with assisted walking for 3 weeks after discharge. Physical rehabilitation then resumes following reassessment of the patient and radiographs if fusion was performed. Standard protocols are prescribed for strengthening and muscle re-education, including ES, passive range of motion, standing exercises, leash walking, proprioceptive and gait training, core muscle strengthening, and aquatic therapy (Figure 34-19). As with most protocols, therapy continues for a 3- to 6-week period and is lengthened if necessary.

Cervical Spondylomyelopathy

Cervical spondylomyelopathy, also called caudal cervical stenotic myelopathy or wobbler syndrome, is a disorder caused by abnormal development of the cervical vertebrae resulting in compression of the spinal cord. Middle-aged (7-9 years) Doberman pinschers and young (1-4 years) giant-breed dogs (Great Danes, mastiffs, rottweilers, Bernese, and Swiss mountain dogs) are most commonly affected.[21] The specific pathologic changes vary, but may include stenosis of the vertebral canal, bony enlargement

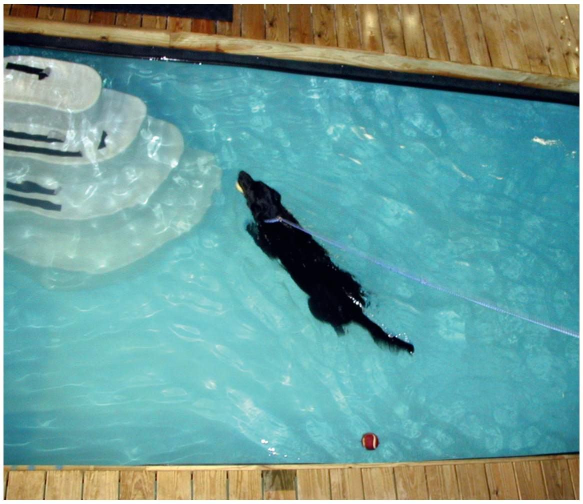

Figure 34-19 Deep water hydrotherapy.

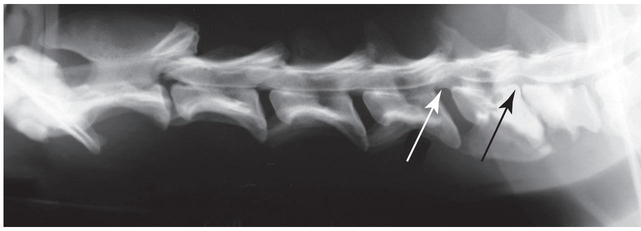

Figure 34-20 Myelography of cervical spondylomyelopathy. There is compression of the spinal segments dorsal to the C5-C6 disk space (*white arrow*). There is a second, less severe lesion dorsal to the *C6-C7* disk space (*black arrow*), which is narrowed.

of the articular facets, disk protrusion, capsular and ligament proliferation, synovial cysts, and vertebral instability. The C5-C6 and C6-C7 vertebrae and disk spaces are most commonly affected.

Clinical signs may be acute or slowly progressive. Mild cases are characterized by subtle ataxia of all limbs, often evident as swaying and increased stride length in the pelvic limbs, with a short-strided gait in the thoracic limbs. Because of the anatomic arrangement of the long spinal tracts to the pelvic limbs, the clinical signs are usually more pronounced in the pelvic limbs. Gait deficits are typically most evident at a slow walk and are less obvious at a fast walk or trot. Additional signs vary and may include low head carriage and neck stiffness. In severe cases there is paresis or paralysis of all limbs. Neck pain is variable, but is more likely in acute cases. The differential diagnosis includes congenital anomalies, trauma, meningomyelitis, discospondylitis, neoplasia, and degenerative, ischemic, and embolic conditions.

Definitive diagnosis is based on spinal imaging. Survey radiographs may be normal or show abnormally shaped vertebra, narrowed disk space(s), and osteoarthritis of the articular facets. Myelography, CT-myelography, or MRI is necessary to fully characterize the specific changes and is essential if surgery is considered (Figure 34-20). Hypothyroidism, von Willebrand disease, and cardiomyopathy are relatively common in middle-age Dobermans, so thyroid testing, buccal mucosal bleeding time, and chest radiographs are indicated in affected Dobermans.

Treatment is based on the severity of the neurologic signs, the results of diagnostic imaging, quality-of-life issues, and the wishes of the owner. Nonsurgical treatment consists primarily of exercise restriction to minimize vigorous activity that would exacerbate the dynamic component of spinal cord compression. Dogs are walked on a leash, and unsupervised activity is avoided. A body harness is worn instead of a neck collar. Antiinflammatory doses of corticosteroids such as prednisone are often helpful. In one study, 54% of patients treated medically improved and 27% remained unchanged over the long term.[22]

Surgery is indicated in patients with substantial neurologic deficits and those that do not adequately respond to nonsurgical treatment. The goals of surgery are to relieve any spinal cord compression and stabilize vertebral instability. The particular technique chosen is based on the specific changes evident on imaging and surgeon preference, and may include ventral slot decompression, dorsal laminectomy, or distraction and fusion of affected vertebrae. Overall 70% to 80% of patients improve after surgery, with a mean time to reach maximum improvement of 2.6 months. About 20% suffer recurrence. There is no clear difference among different surgical techniques.[22,23] Aftercare is a major part of the treatment, requiring significant intervention and effort of the caregiver.

Physical Rehabilitation for Cervical Spondylomyelopathy

Physical rehabilitation plays a significant role in patient care. Modalities directed toward decreasing muscle spasms, such as US and ES, passive range of motion of limbs, and support with a soft sling to provide assisted standing and weight bearing, may be used in nonambulatory patients. Assisted sling walking and assisted aquatic therapy are useful to help patients regain unassisted ambulation. Neuromuscular ES for muscle building and re-education is performed in selected cases. As ambulation improves and strength is gained, unassisted deep-water and underwater treadmill therapy may be started. Gait training and strengthening exercises are incrementally implemented. Modalities used must be compatible with the patient's neurologic status and the stage of wound healing in the postoperative period. Extended time in rehabilitation is often necessary for proper management of these cases because of the difficulties in managing large and giant breeds in a home environment.

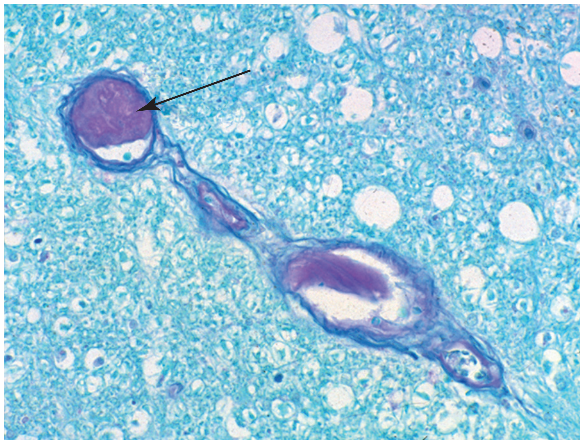

Figure 34-21 Photomicrograph of fibrocartilaginous embolic myelopathy. There is a piece of fibrocartilage (*arrow*) occluding one of the vessels in the spinal cord. There is necrosis of the adjacent portion of the spinal cord (hematoxylin eosin with PAS stain).

Fibrocartilaginous Embolic Myelopathy

Fibrocartilaginous embolic myelopathy (FCEM) is an acute infarction of the spinal cord caused by a vascular embolus of fibrocartilage, probably originating from the intervertebral disk (Figure 34-21). Adult large- or giant-breed dogs and miniature schnauzers are most commonly affected. This disease is less common in small dogs, chondrodystrophic dogs, and cats.

The onset is sudden and often associated with activity such as running or playing. Neurologic deficits rarely progress beyond the first few hours. Although patients sometimes yelp as if in pain at the onset of signs, spinal pain is rarely evident by the time of examination. This lack of focal spinal pain is helpful in ruling out other causes of acute spinal diseases, such as fracture/luxation and disk extrusion. Any region of the spinal cord may be affected, and the spinal cord segments involved dictate the specific neurologic deficits. Ataxia, paresis, or paralysis may affect the pelvic limbs or all limbs. Asymmetric or unilateral deficits are common. Lower or upper motor neuron deficits may result. In severe cases there is loss of deep pain perception caudal to the lesion.[24,25]

Diagnosis is based on the clinical features and exclusion of other causes. Essential features of FCEM are the signalment, sudden onset, nonprogressive course, lack of focal spinal pain, and often the asymmetry of deficits. The differential diagnosis includes intervertebral disk disease, trauma, neoplasia, and inflammatory disease. Radiographs and myelography are typically normal, although myelography occasionally shows focal spinal cord swelling.[24,25] Cerebrospinal fluid analysis is nonspecific. Magnetic resonance imaging is the best imaging modality and usually shows signal changes suggestive of focal spinal cord infarction.[26]

There is no specific treatment for this condition. Corticosteroids are sometimes used although there is little evidence that any drug is beneficial. Nursing care and physical rehabilitation play an essential role in promoting recovery and preventing complications. About 85% of patients recover, depending on the severity of the deficits and owner's commitment.[26]

Physical Rehabilitation for Fibrocartilaginous Embolic Myelopathy

Modalities and techniques used are similar to those used for patients with intervertebral disk disease. Rehabilitation techniques vary depending on the location of the lesion and the degree of damage to the spinal cord. Depending on the severity of the condition, the patient may have increased muscle tone and spasticity, or decreased muscle tone and flaccidity. In general, ES, passive range of motion of affected joints, aquatic therapy, and assisted ambulation are early modalities that may be used. Continued ES, assisted sling walking, deep water aquatic therapy, and gait and strength training are performed as the patient improves.

Nonambulatory patients require increased nursing care, including appropriate bladder care, assistance with adequate nutrition and fluid intake, and appropriate monitoring to ensure cleanliness and comfort. Patients should be repositioned every 4 to 6 hours to prevent decubital sores. A positioning protocol should be initiated immediately. If increased tone to the limb(s) is present, at-risk muscles are placed in elongated positions. Passive range of motion to joints and muscles at risk of developing contracture and adaptive shortening is performed regularly, typically three to six times daily. Passive range of motion or active assisted range of motion and stretching using the body weight and movement of the body in closed-chain activities can be effective. Slow rhythmic bouncing on a therapy ball assists in decreasing hypertonia and allows for more effective muscle stretch and more normal movement patterns.

If the patient is unable to sit or stand independently, assistance to these positions, with proper joint alignment, should be promptly initiated. Placing the patient over a small therapy ball with the feet still in contact with the floor assists in maintaining a standing position (see Figure 34-12). It is important to provide weight bearing on the affected limbs while maintaining a standing position. Weight shifting in functional planes of movement also may be initiated in this position. If working only with the pelvic limbs, the torso and front limbs may be placed on the ball while the pelvic limbs are worked in a weight-bearing position.

Firm, deep pressure through manual contacts activates the tactile receptors, neuromuscular proprioceptive pathways, and sensory awareness. Proper hand placement provides security and support to unstable body segments and may stimulate contraction of the muscles under the hands of the therapist (Figure 34-22). Applying approximation

Figure 34-22 Assisted standing exercise.

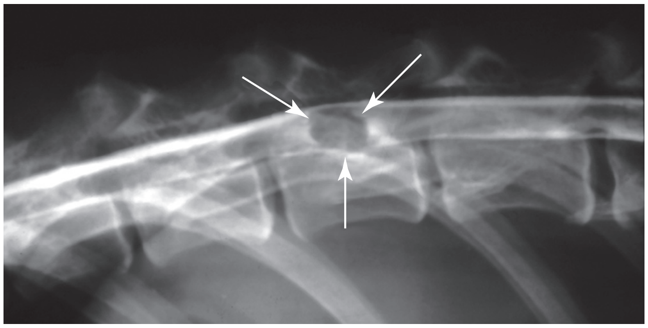

Figure 34-23 Myelography showing a spinal tumor (meningioma) at the level of L1. There is a well-defined filling defect (*arrows*) outlining the mass.

and compression joint mobilization techniques facilitate postural extensor and stabilizers and enhances body awareness. Approximation of the affected joints can be achieved by bouncing the patient while it is sitting on a therapy ball or weight bearing over a ball, and through manual compression of the joints. Providing intermittent pressure through the pelvis and legs (bouncing) stimulates approximation of affected joints in patients with pelvic limb dysfunction that are able to assume a standing position.

Deep water hydrotherapy is also used to stimulate muscle activity and motor learning (see Figure 34-19). Depending on the functional level of the patient, life preservers or manual assistance in the pool may be necessary to ensure safety (see Figure 34-6). This assistance also allows the therapist to more easily provide reciprocal movements to the affected limbs. Assisting the patient to stand in shallow water or weight bearing on the pelvic limbs with the forelimbs elevated are also useful treatments in the pool.

Assisted ambulation with manual reciprocal placement of the patient's feet may stimulate motor memories and initiate motor learning. Sling or tail walking aids ambulation. As the patient improves, the use of an underwater treadmill or land treadmill further benefits strengthening and gait training. Standard 3-week rehabilitation protocols may be lengthened in those patients experiencing a slower recovery.

Neoplasia

Spinal tumors are classified by their location as extradural, subarachnoid (intradural-extramedullary), and intramedullary. Extradural neoplasia is most common and includes osteosarcoma, multiple myeloma, fibrosarcoma, chondrosarcoma, hemangiosarcoma, and lymphosarcoma. Extradural lymphosarcoma is the most common spinal tumor in cats and may be secondary to infection with the feline leukemia virus. Subarachnoid tumors include nerve sheath tumors, meningioma, lymphosarcoma, and neuroepithelioma. Intramedullary tumors include glioma, lymphosarcoma, and hemangiosarcoma. Metastatic tumors, including metastatic carcinomas and sarcomas, may occur in any location. Spinal neoplasia is more common in middle-aged or older animals. However, neuroepithelioma affects young, large breed dogs, and lymphoma may affect a cat at any age.

Clinical signs are typically insidious and progressive but can be acute. Focal spinal pain is the most common initial sign. Later ataxia and paresis may develop as the spinal cord becomes progressively affected. Systemic signs, such as weight loss, may occur with metastatic cancer. Nerve sheath tumors often present as chronic lameness of one limb that can be difficult to differentiate from orthopedic disease. Eventually, muscle atrophy and neurologic deficits develop. Intramedullary tumors may present with progressive neurologic signs without evidence of pain.

Diagnosis is based on clinical features, and diagnostic imaging. If neoplasia is suspected, chest radiographs are indicated to screen for metastasis. Spinal radiographs may show abnormalities in cases of vertebral tumors, although initial changes may be subtle. Myelography is helpful in identifying spinal cord compression (Figure 34-23). Computed tomography, CT-myelography, and MRI are more sensitive. Definitive diagnosis often requires surgical exploration and biopsy.

Medical management may provide temporary relief of pain and weakness to maintain a reasonable quality of life for a short period of time. Generally this consists of judicious activity, analgesics, and corticosteroids. Chemotherapy may be appropriate for specific tumor types. Definitive treatment usually consists of surgery. The goal is to obtain tissue for histopathologic diagnosis, relieve compression on the spinal cord, decrease the tumor volume before radiation therapy, or completely remove the tumor, depending on the location, size, and type of tumor. Radiation therapy may be used as a primary treatment or as an adjunct to surgery, depending on tumor type. The prognosis depends on the tumor type.

Physical Rehabilitation for Neoplasia of the Spinal Column

Postoperative physical rehabilitation protocols are generally similar to those used in the rehabilitation of patients after intervertebral disk decompression. One essential difference, however, is that protocols must generally be stepped down rather than stepped up as the disease progresses and the condition and ability of the patient diminish.

Trauma

Major trauma can result in various head and spinal lesions, depending on the anatomic location. A thorough history usually documents the presence and mechanism of trauma, although the cause is sometimes unknown or surmised by concurrent findings if the accident is not witnessed. Automobile accidents, gunshot wounds, falls, and dog fight injuries are common. Spinal injuries may consist of vertebral luxation, vertebral fracture, vertebral fracture/luxation, or traumatic IVD herniation.

In any trauma patient, full assessment of all body systems is essential to diagnose and treat shock and any other injuries. A neurologic examination is performed to localize the injury and determine the severity of the deficits. It is vital that the examination be carefully performed, minimizing movement of the patient to avoid further damage to the spinal cord as a result of instability of the spinal column. Assessment of mentation, cranial nerves, voluntary movement, and spinal reflexes, as well as head and spine palpation, is performed, but moving the patient to test gait and postural reactions is avoided until unstable spinal injuries are ruled out with radiographs. The presence or absence of deep pain caudal to the injury is the most important prognostic sign; lack of deep pain sensation after spinal trauma carries a poor prognosis for neurologic recovery.

Following initial assessment and stabilization, appropriate analgesics are administered. Methylprednisolone (30 mg/kg IV, then 1 hour later 5.4 mg/kg per hour IV infusion for 24 to 48 hours) may improve the chance of recovery from severe spinal cord injury if started within 8 hours of injury. However, conclusions from human clinical trials are controversial and there are no published controlled studies in veterinary patients.[27] A lateral-view radiograph of the affected region of the spine is obtained when the patient is stable. If dorsoventral views are necessary, a horizontal beam view is safest. The purpose of radiography is to confirm the localization, classify any fracture/luxation as stable or unstable, and try to determine if there is any persistent spinal cord compression that may require surgery. The two components of the spine that maintain stability are (1) the dorsal compartment, consisting of the articular facets and lamina, and (2) the ventral compartment, consisting of the vertebral bodies and intervertebral disk. Damage to one of these components usually does not result in severe instability, whereas damage to both the dorsal and ventral components usually requires fixation to prevent further displacement. Myelography or CT is indicated if survey radiographs are normal or inconclusive. Routine chest and abdominal films also may be indicated to determine other body systems affected by the trauma.

The decision to treat the patient is based on the severity of the neurologic deficits, the type of vertebral injury, the severity of any concurrent injuries, and the owner's understanding of the risk, prognosis, and expense of the injury and treatment. Animals with mild neurologic deficits caused by cervical injuries or stable thoracolumbar injuries often recover with nonsurgical therapy. This consists of 4 to 6 weeks of complete cage rest, with very cautious movement of the animal to assist it with posturing to urinate and defecate. Weight bearing is generally not encouraged, unless the vertebral column is stable. Sometimes an external splint is used in conjunction with cage rest.[28]

Indications for surgery include (1) unstable fracture/luxation, usually evident on imaging by disruption of the dorsal and ventral components of the vertebrae; (2) persistent compression of the vertebral canal, evident as a substantially displaced fracture/luxation or extradural compression of the spinal cord on myelography, CT, or MRI; and (3) progressive deterioration in neurologic status despite conservative treatment, usually owing to an unstable fracture/luxation or progressive spinal cord compression from hemorrhage or disk extrusion.[28]

Surgery usually involves decompression of the spinal cord and fixation of unstable vertebrae. The choice of fixation technique depends on the size of the animal, the type and location of the injury, and the surgeon's preference and experience. In general, dorsal techniques (spinal staples, dorsal spinal plates, and modified spinal instrumentation) are indicated if the ventral components of the vertebrae are intact. Ventral techniques (vertebral body plates, pins/screws, and bone cement) are stronger and can be used with dorsal and ventral component injuries (Figure 34-24). A laminectomy to decompress the spinal cord from intervertebral disk material, ligament, or bone may be necessary before fixation. External splints of light casting material or moldable thermoplastic materials provide some stability and are most applicable to thoracic, cranial lumbar, and cervical regions. Splints may create sores and have the potential for creating complicated wounds that require additional care. Splints may also be used as an adjunct to internal fixation.[28]

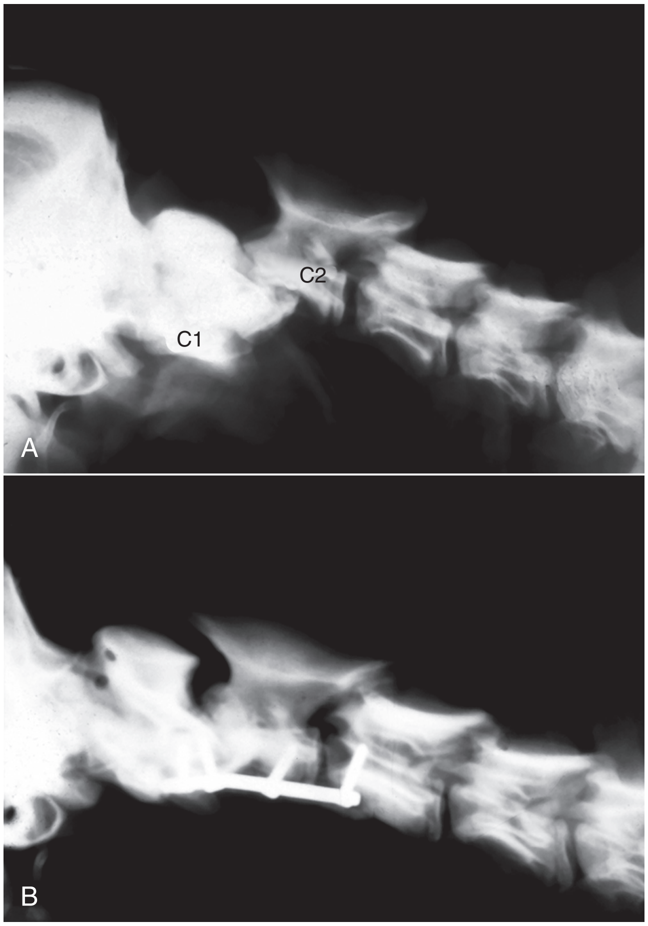

Figure 34-24 Surgery for a spinal fracture/luxation. **A,** There is a fracture of C2 with dorsal displacement of C2 relative to C1. **B,** Postoperative view. The injury has been repaired with a plate and screws affixed to the ventral aspect of C1, C2, and C3.

Physical Rehabilitation for Traumatic Injuries

Physical rehabilitation protocols and modalities are similar to those used for rehabilitation of lumbosacral decompression/stabilization and cervical spondylomyelopathy patients. Wound care is discussed in detail in chapter 36 for those patients that develop cast wounds. Extended care and therapy beyond the standard 3-week rehabilitation protocol may be required in some cases.

Peripheral Nerve Injury

Common causes of peripheral nerve injury include fractures (e.g., femoral fracture that damages the sciatic nerve), intramuscular injection (usually affecting the sciatic nerve), traumatic brachial plexus nerve root avulsion, and poor surgical technique. Vascular injuries can also occur, the most common of which is iliac thrombosis in cats causing a distal sciatic neuropathy, but thrombosis of the brachial artery can also cause thoracic limb monoparesis. Axons in peripheral nerves regenerate at rates as fast as 1 mm a day. Nerves must be in a Schwann cell environment for this regeneration to occur. Peripheral nerve injuries have three levels of severity. In neurapraxia, axonal conduction is lost without disruption of the axon. This injury usually results from compression, transient ischemia, or blunt trauma. Loss of conduction may be a result of myelin damage or insufficient energy to maintain axonal resting potential. In axonotmesis the axon integrity is lost, but the endoneurium and Schwann cell sheath it lies within are still intact, providing the opportunity for regeneration back to the correct target. Successful regeneration may occur, particularly if the axon is damaged close to its target. In neurotmesis the entire structure of the nerve is disrupted. The axon has the ability to regenerate but needs to find a Schwann cell sheath to do so, making it much more difficult.

Peripheral nerve lesions can cause unpleasant abnormal sensations (paresthesia) and hyperesthesia, both of which can result in self-mutilation. Another sequel to denervation of a muscle is severe muscle atrophy, which over time may lead to muscle contracture and, in growing animals, to skeletal deformities.

Localization of the lesion is based on the neurologic examination, which indicates the pattern of muscle denervation and sensory loss. It is useful to refer to references depicting the cutaneous sensory zones of peripheral nerves.[29,30] Severity of the lesion is determined by assessing the level of motor function and assessing for deep pain sensation. Electrophysiologic evaluation of the muscles and nerves using electromyography and nerve conduction velocity studies allows a more detailed description of the severity and course of the injury. Muscles that are completely denervated develop spontaneous electrical activity when at rest, although such changes do not appear for at least a week after denervation of a muscle. Nerve conduction studies should be interpreted with care. Immediately after an injury, conduction may be lost across the site of injury, whereas the distal portion of the disrupted nerve can continue conducting for a period of hours to days. As nerves regenerate and sprout to innervate denervated muscles, the size of motor units increases; therefore the size of motor unit potentials on the EMG increases.[31,32]

As a rule neurotmesis carries a poor prognosis unless immediate surgical intervention to reconnect the severed nerve occurs. Animals with axonotmesis or neurapraxia carry a better prognosis. Neurapraxia usually reverses within 2 weeks of injury, although damage to myelin slows recovery to 4 to 6 weeks.

The recovery from axonotmesis is governed by the proximity of the injury from the target muscle, the severity of muscle atrophy, and the development of contractures. If the damage has occurred far from the target muscle (e.g., at the brachial plexus), by the time the axon has regrown, severe muscle atrophy and contractures could limit recovery.

Brachial plexus injuries tend to involve the caudal two thirds of the plexus (radial, median, ulnar, and lateral thoracic nerves and the sympathetic innervation of the head) or the complete plexus, although cranial plexus injuries have been reported.[31] It is easy to be misled when evaluating animals with caudal plexus injuries, because there is preservation of musculocutaneous function and elbow flexion. This function is not useful for recovery of the ability to bear weight and should not be used to determine the prognosis. Instead, it is important to test deep pain perception, particularly in the lateral digit.[33] The absence of deep pain in this digit implies severe radial nerve injury. If it does not reappear within 2 weeks of injury, the prognosis for recovery of useful motor function in that limb is guarded.

In peripheral nerve inures, sensation must be considered. Dogs experiencing anesthesia, paresthesia and hyperesthesia may injure their limbs by not recognizing when the limb may be caught on an object, dragging the limb and causing damage, or by self-mutilation caused by the unpleasant sensations.

Rehabilitation for Peripheral Nerve Injury

The goals of a rehabilitation program for patients with lower motor neuron injury include restoring and maintaining joint range of motion, improving muscle strength, restoring neuromuscular function, and preventing self-mutilation and trauma to the affected limb. The lack of spinal reflexes and corresponding muscle tone in these patients poses a challenge to their rehabilitation, and emphasis is placed on restoration of muscle and joint function.

Because of the dysfunction of the spinal reflex arc, passive exercises are performed until a near-normal gait is established. Passive range of motion and stretching are used in the same manner as in patients with other neurologic problems. Patients with lower motor neuron deficits may benefit from stretching of affected and antagonist muscles. Loss of tone to antagonist muscle groups predisposes patients to joint contractures. Massage of a mildly contracted muscle group may also be beneficial in restoring its function and should be performed two to three times per day after warming the region.

In patients with a sciatic nerve deficit, elicitation of a flexor withdrawal reflex may not be possible. Nevertheless, progress should be monitored by serial evaluations of the spinal reflex arcs. In patients with weak or intact flexor withdrawal reflexes, stimulation of the flexor reflex helps to improve muscle tone and neuromuscular coordination. Patients with femoral nerve injury require much assistance to maintain normal posture. A balance ball (Swiss ball) may be used to support the trunk while slowly lowering the pelvic limbs to the ground. The patient's hind limbs are then gently raised (enough to lift the toes off the ground) and lowered, such that the animal is required to support its body weight as the hind limbs are lowered to the ground.

Patients with mild radial nerve deficits benefit from being challenged to bear weight on their forelimbs. Patients who lack elbow or carpal extension (e.g., brachial plexus avulsion) have great difficulty or are unable to perform this activity until some extensor muscle tone is present. The exercise is performed by placing the patient in a standing position while supporting the trunk and thoracic limbs. With the animal's front feet placed squarely on the ground, the amount of support is gradually reduced. When the patient starts to collapse in the thoracic limbs, support is provided to the animal and it is returned to a standing position. A balance ball or custom orthotics may similarly be used to support the patient. Orthotics are described in more detail in Chapter 17. The exercise is repeated 5 times, two to three times per day.

Active exercises are designed to improve muscle strength, neuromuscular balance, and coordination in patients who have at least some voluntary movement of their limbs. Patients with peripheral nerve disease affecting more than one limb may not be able to do some of these activities. Patients with sciatic nerve deficits can often perform sit-to-stand exercises because these exercises require active stifle extension but only passive stifle and tarsal flexion. Balance and coordination exercises benefit patients with peripheral nerve injuries and are performed as described previously.

The application of neuromuscular electrical stimulation in patients with peripheral nerve disease may delay the onset of neurogenic muscle atrophy and recondition the affected muscles.[34] When a muscle is completely denervated, electrical muscle stimulation is the modality of choice. Affected muscle groups should be stimulated once a day for 15 minutes each.

Discospondylitis

Discospondylitis is an infection of the intervertebral disk and adjacent vertebral bodies. It is usually caused by hematogenous spread of organisms from sites elsewhere in the body, such as the urinary tract, skin, or mouth. Penetrating wounds, surgery, or plant material migration can cause direct infection of the disk space or vertebra. In certain geographic regions migrating plant material such as grass awns is a relatively common cause of vertebral osteomyelitis and discospondylitis of the L2-L4 vertebrae. *Staphylococcus intermedius* is the most common etiologic agent identified in dogs with discospondylitis. Other frequently documented pathogens include *Streptococcus* spp., *Brucella canis*, and *Escherichia coli*, but virtually any bacterial agent can be causative. Some infections are caused by fungal organisms, including *Aspergillus*, *Paecilomyces*, and *Coccidioides immitis*. Vertebral infections caused by grass awn migration are often associated with mixed infections, with *Actinomyces* being a common isolate.[35]

Discospondylitis most commonly affects large, middle-aged dogs. Small dogs and chondrodystrophic breeds are uncommonly affected, and the disease is rare in cats. The most common initial sign is spinal pain, which varies in intensity from mild to extreme. The duration of signs noticed by the client usually ranges from several days to several weeks. Neurologic deficits ranging from ataxia to paralysis are seen in some patients, although usually not as the initial abnormality. Signs suggestive of systemic illness, such as fever, lethargy, depression, anorexia, and weight loss occur in about 30% of affected dogs. Concurrent urinary tract infection, including prostatitis in male dogs, is common.[35]

Diagnosis is based on clinical features and diagnostic imaging. Radiographs of the involved region are usually diagnostic, although they may be normal early in the course of disease. The earliest radiographic sign is subtle irregularity of the vertebral end plates. As the infection progresses the erosion of the end plate becomes more pronounced, and there are osteolytic and osteoproductive changes of the intervertebral disk space, vertebral end plates, and in severe cases, the entire vertebral body (Figure 34-25). Computed tomography and MRI are more sensitive and are helpful when radiographs are normal or inconclusive. Myelography or MRI is indicated only in patients with substantial neurologic deficits when decompressive surgery is considered. Complete blood count, urine culture and sensitivity, blood cultures, and serology for brucellosis are performed in patients with radiographic evidence of vertebral infection. The differential diagnosis includes chronic intervertebral disk disease, inflammatory disease, and neoplasia.

Initial treatment consists of antibiotics, cage rest, and analgesics. Selection of antibiotics is based on culture and sensitivity of blood and urine and serology for *B. canis*. Pending these results or if these tests are negative, treatment is started with a bactericidal antibiotic effective against beta-lactamase–producing *Staphylococcus*. First-generation cephalosporins are effective against the majority of isolates from affected dogs and are a good choice for initial therapy. Clindamycin and amoxicillin with clavulanic acid are also usually effective. Patients with fever, neurologic deficits, or rapidly progressing signs are treated with intravenous antibiotics for 3 to 5 days, followed by a course of oral antibiotics. Initial oral therapy is appropriate in patients with mild, slowly progressing spinal pain.[35]

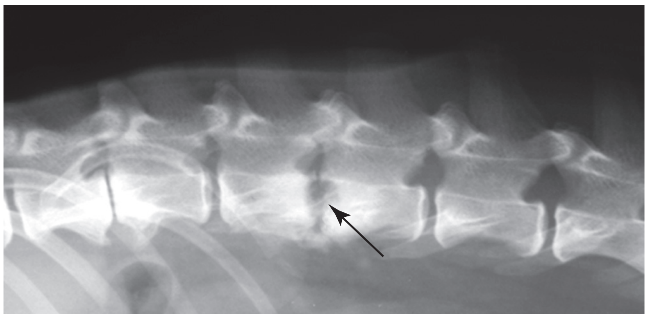

Figure 34-25 Discospondylitis. The radiograph shows lysis of the vertebral end plates (*arrow*) adjacent to the L2-L3 disk space.

Clinical signs such as spinal pain and fever usually resolve within 5 days of starting effective therapy, although neurologic deficits usually resolve more slowly. Patients should be treated for at least 8 weeks in an attempt to completely clear the infection and prevent relapse. Patients that fail to improve within 5 days of starting treatment are reassessed. Options include treating with a different antibiotic or culture of the affected disk space by needle aspiration, guided by fluoroscopy or CT.[36] Most animals recover with appropriate medical treatment. Surgery is rarely indicated to obtain biopsy when there is no response to initial therapy, or to decompress the spinal cord in patients with substantial neurologic deficits, evidence of spinal cord compression on imaging, and no response to medical therapy.

Physical Rehabilitation for Discospondylitis

Physical rehabilitation is indicated in more severely affected patients after a positive response to medical management is achieved. Protocols used are similar to those used in dogs with IVD, specifically aquatic therapy, and strength and gait training. Particular care should be exercised because affected patients generally have significant weakness of the affected bones of the vertebral column.

Atlantoaxial Luxation

The anatomy of the atlas (C1) and the axis (C2) is distinctly different from the rest of the vertebral column. There is no intervertebral disk and very little flexion at this site. The anatomical relationship between C1 and C2 is primarily maintained by ligaments. Atlantoaxial luxation occurs when instability of the joint allows C2 to luxate dorsally, relative to C1, and compress the spinal cord. Associated lesions can include congenital absence of the dens, fracture of the dens, and absence or rupture of atlantoaxial ligaments.

Atlantoaxial luxation is most common in toy-breed dogs with malformation of the dens or ligaments. It can occur in any animal as a consequence of trauma. In dogs with congenital instability of the atlantoaxial joint, clinical signs usually occur within the first few years of life and are often precipitated by relatively minor trauma. Yorkshire terriers, Chihuahuas, miniature and toy poodles, Pomeranians, and Pekingese dogs are most commonly affected. Clinical signs include neck pain and gait dysfunction, ranging from ataxia to tetraplegia, with normal or exaggerated reflexes. Severe cases may be fatal due to

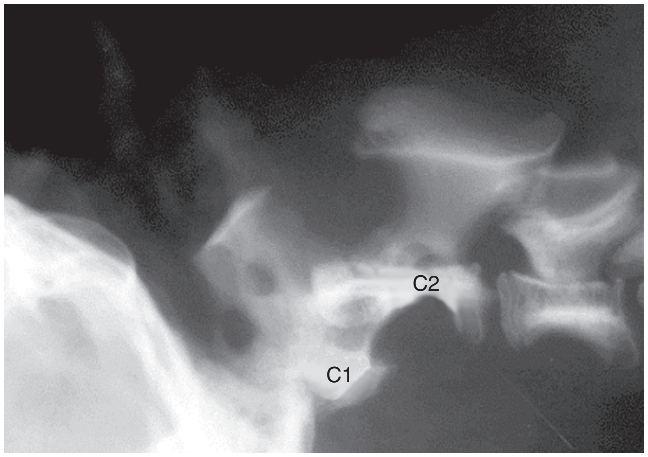

Figure 34-26 Atlantoaxial luxation. The radiograph shows dorsal displacement of C2 relative to C1.

respiratory paralysis. It is critical to avoid flexion of the neck because this exacerbates spinal cord compression. The differential diagnosis includes vertebral fracture or luxation, intervertebral disk disease, discospondylitis, neoplasia, and meningomyelitis.

The diagnosis is based on history, physical and neurologic examination, and radiographs. Typical radiographic findings include dorsal displacement of C2 relative to C1 and an increased space between the arch of C1 and spine of C2 on the lateral view (Figure 34-26). The size and shape of the dens are best evaluated on a ventrodorsal view. The dens may be normal, hypoplastic, aplastic, or dorsally angulated. If survey radiographs are equivocal, the atlantoaxial joint may be slightly flexed during fluoroscopy to assess stability. CT or MRI is also useful if the diagnosis is uncertain, or to identify coexisting lesions. Cerebrospinal fluid analysis is helpful in identifying or ruling out inflammatory disease. Lumbar puncture for fluid collection avoids flexion of the cervical spine that is required for collection of fluid from the cerebellomedullary cistern.

Nonsurgical treatment is indicated in patients with mild signs. This consists of cage rest and application of a neck brace that prevents neck flexion for 6 weeks. Analgesics are administered as needed. Conservative treatment may be successful in patients with traumatic luxations, but dogs with congenital instability can suffer relapse because of the inherent instability. Surgery to reduce and stabilize the atlantoaxial joint is the definitive treatment. Doral techniques achieve fixation by securing the spinous process of the C2 to the dorsal arch of C1 using the nuchal ligament, orthopedic wire, nonmetallic suture, or an implantable device (Kishigami atlantoaxial tension band). Cross-pinning of the atlas and axis with polymethylmethacrylate has also been described. Ventral techniques can be performed using transarticular screws or pins, with or without polymethylmethacrylate, or a plate. Implants may also be placed into the pedicle of the atlas and the body of the axis. The articular cartilage may be removed and bone grafts placed to encourage fusion. A theoretical advantage of ventral techniques is bony fusion although incomplete bony fusion is apparent on follow-up radiographs in many patients with a good outcome.[37] Potential complications include implant failure, worsening of neurologic deficits, and upper respiratory problems associated with the ventral surgical approach. Successful outcome is reported in 62% to 91% of patients. Older age (greater than 24 months) at onset of signs, longer duration of clinical signs (greater than 10 months), and more severe neurologic deficits increase the risk of poor outcome.[37]

Physical Rehabilitation for Atlantoaxial Subluxation

Postoperatively, a cervical splint fashioned from light casting material may be applied. Periodic removal allows assessment of underlying tissues. External coaptation is continued for 4 weeks along with strict confinement. Physical rehabilitation begins after cast removal. Whirlpool aquatic therapy combined with ES and therapeutic US is appropriate in the early phases of rehabilitation. Assisted aquatic therapy is beneficial until ambulation begins. Strengthening and gait training exercises and underwater treadmill aquatic therapy may be incrementally added.

Degenerative Myelopathy

This is a slowly progressive degenerative disease of the central nervous system that most severely affects the thoracic spinal cord segments. It is inherited as an autosomal recessive trait. The cause is a mutation in the superoxide dismutase 1 (SOD1) gene, similar to the familial form of amyotrophic lateral sclerosis in human patients (Lou Gehrig's disease).[38]

Degenerative myelopathy affects older dogs, usually more than 8 years of age. German shepherd dogs, Rhodesian ridgebacks, Pembroke and Cardigan Welsh corgis, boxers and Chesapeake Bay retrievers, are most commonly affected, although cases have been documented in a number of other large-breed dogs and cross breeds.[39]

There is an insidious onset of progressive ataxia and paresis in the pelvic limbs. This is manifested as swaying, staggering, and scuffing of the rear limbs, often worse on one side. Spinal reflexes in the pelvic limbs are normal to exaggerated. An essential feature is that this disease does not cause spinal pain. Most patients become nonambulatory within 6 to 9 months, at which point many clients elect euthanasia. Otherwise the signs eventually progress to loss of reflexes in the pelvic limbs, paresis of the thoracic limbs, urinary and fecal incontinence, and brainstem involvement, such as dysphagia.[39]

Diagnosis is based on clinical features, imaging, and spinal fluid analysis to rule out other causes, and DNA testing to detect the SOD1 mutation. Affected dogs have two copies of the abnormal gene, one inherited from the mother and one inherited from the father. Dogs that test positive for both copies of the abnormal gene are at risk of developing the disease and will pass the mutation to offspring. Dogs with one copy of the abnormal gene do not develop the disease but can pass the trait on to their offspring. Dogs with two copies of the normal gene do not develop the disease and cannot pass on the trait.[39]

Other spinal cord diseases can cause similar or identical signs, including intervertebral disk protrusion, spinal cord tumor, and inflammatory diseases. It is important to exclude these diseases on the basis of imaging tests such as MRI and analysis of spinal fluid.

There is no medication that has been shown to be effective for degenerative myelopathy in randomized controlled studies. Steroids and other antiinflammatory drugs have been used but are generally not helpful. A study evaluating combination therapy with aminocaproic acid, *N*-acetylcysteine with vitamins B, C, and E failed to detect a benefit.[40] Physical rehabilitation may improve the quality of life of affected dogs and improve survival time. A nonrandomized study found patients with extensive physical rehabilitation had significantly longer survival times (mean 255 days) than dogs that received moderate (mean 130 days) or no physical rehabilitation (mean 55 days).[41] This is likely due to dogs maintaining their mobility longer with physical rehabilitation as loss of mobility is a main factor in euthanasia. Common rehabilitation includes range of motion and therapeutic exercise, aquatic therapy, and maintaining function as long as possible.

Miscellaneous Conditions

Miscellaneous diseases creating neurologic dysfunction that are treatable and have a reasonable prognosis may benefit from physical rehabilitation. These include immune-mediated polyradiculoneuritis (Coon Hound paralysis), congenital vertebral anomalies and arachnoid cysts. Because of the uncommon clinical presentation of these conditions and the lack of experience in rehabilitating these conditions, the use and success of specific protocols are unknown. Adherence to therapeutic principles and selecting protocols used for similar diseases should provide a sound basis for rehabilitation of these patients.

Summary

The rehabilitation of dogs with neurologic disease involves a combination of nursing care, active and passive exercise, functional activities, and therapeutic modalities. The foundation for maximizing the patient's functional recovery is cooperation and participation of the patient, the owner, and the therapist.

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35 Physical Rehabilitation for Geriatric and Arthritic Patients

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Geriatric and arthritic patients have specific medical and rehabilitation needs linked to the increased prevalence of chronic conditions and their progressive loss of mobility. In fact, geriatric patients can be a significant component of a rehabilitation practice. Aging triggers changes in a wide range of tissues. Also, the incidence and severity of chronic diseases increase over time. The life span of canine companions is increasing in part because of medical and surgical advances in veterinary medicine. As a consequence geriatric and arthritic patients are often candidates for rehabilitation, requiring a thorough assessment and potentially complex treatments. Geriatric and arthritic patients should also be reassessed on a regular basis to detect changes in their conditions and to design and deliver the most accurate and beneficial plan of care. This chapter discusses the effects of aging, medical considerations and common neurologic and musculoskeletal conditions in the aging dog, quality of life issues, pain management in geriatric and arthritic patients, nutritional considerations, and general management of the arthritic patient.

Aging in Dogs

The term *geriatric* relates to elderly people. The World Health Organization defines middle-age as being 45 to 59 years, elderly as being 60 to 74 years, and aged as being 75 years and older.[1] No such groupings have been proposed in dogs, although Table 35-1 describes the ages at which dogs are considered "geriatric" and are likely to develop diseases associated with aging.[2] Aging itself is not a disease. Multiple factors, including genetics, environment, and nutrition, play a role in the aging process. For example smaller dog breeds live longer than larger breeds, and mixed-breed dogs have a longer life expectancy than purebred dogs. Obese animals have a shorter life span than non-obese animals.[3] Dogs that are neutered live longer than those that are not neutered. Pets that live inside live longer than those that are housed outside.[4]

There are multiple changes that occur as a result of the aging process in dogs, as outlined in Box 35-1.

Most of these conditions are chronic (endocrinopathies, hepatic disease, renal disease, cardiac insufficiency) and require lifelong treatment and sustained owner compliance. It is important for the therapist to understand the effects these diseases have on function and activities of daily living. For example, hypothyroidism may cause lethargy and easily could be mistaken for arthritis and inactivity. It is important for the owner to make regular appointments with their veterinarian for patient monitoring and modification of the treatment plan, as necessary.[4] The changes associated with aging are progressive and are hastened by stress, the environment, genetic factors, malnutrition, and lack of activity.

Medical Considerations

The effects of aging have been documented in human beings,[5] and similar effects are seen in dogs. Quite often, owners of geriatric dogs attribute specific changes in behavior and lifestyle, such as a decreased appetite, changes in sleeping patterns, or decreased activity, to their pet becoming arthritic or being old. Although a component of these changes could be the aging process, the presence of concurrent neurologic, musculoskeletal, or metabolic diseases should also be considered. In geriatric dogs, the diagnosis and treatment of systemic diseases are necessary for the success of a rehabilitation program aimed at managing a disorder such as osteoarthritis (OA). The increased likelihood of multiple conditions being present in a geriatric dog warrants a thorough medical evaluation before a rehabilitation program is designed. A complete medical history, including previous medical diagnoses and treatments, may help determine the origin of the patient's primary problem and identify any previous or concurrent diseases that may affect the success of rehabilitation. In conjunction with the history, performing a thorough physical examination and obtaining hematologic information may confirm the presence of concurrent diseases.

Multiple disease processes in older animals, such as those listed in Table 35-2, can cause nonspecific signs that mimic or exacerbate the signs of OA or aging. These signs include weakness, lethargy, and exercise intolerance. Treatment of concomitant diseases before the implementation of a rehabilitation program allows the patient to be in the best condition to appropriately respond to rehabilitation

Table 35-1 Typical Ages at Which Various-Sized Dogs May Be Considered Geriatric

	Weight	Age (Mean ± Standard Deviation)
Small dogs	0-20 lb	11.48 ± 1.86
Medium dogs	21-50 lb	10.19 ± 1.56
Large dogs	51-90 lb	8.85 ± 1.38
Giant dogs	>90 lb	7.46 ± 1.27

Box 35-1 Effects of Aging in Dogs

Metabolic Effects

Decreased metabolic rate
Decreased immune competence
Decreased phagocytosis and chemotaxis (decreased ability to ward off infections)
Autoantibodies and autoimmune diseases may develop

Physical Effects

Percentage of body weight represented by fat increases
Skin becomes thickened, hyperpigmented, and less elastic
Footpads hyperkeratinize and claws become brittle
Muscle, bone, and cartilage mass are lost, with possible development of arthritis
Lungs lose elasticity, fibrosis occurs, and pulmonary secretions become more viscous
Pulmonary vital capacity is diminished
Cough reflex and expiratory capacity decrease
Urinary incontinence frequently develops
Cardiac output may decrease
Number of cells in the nervous system decreases

and allows the clinician to provide the most accurate prognosis. If the concurrent disease requires long-term treatment, frequent adjustments of the medical therapy and the rehabilitation program may be required to reach the goals of each treatment. For example some of the most common systemic diseases in older animals are endocrine disorders and neoplasia.[6] Those conditions that have associated neurologic and musculoskeletal involvement, such as hyperadrenocorticism, hypothyroidism, hyperparathyroidism, osteosarcoma, and insulinoma interfere with the implementation of a successful rehabilitation program and accurate assessment of the patient's progression during the program. Understanding the pathophysiologic mechanisms is important not only for the successful treatment of the concurrent disease but also for the development of a rehabilitation plan tailored to the needs and weaknesses of the individual patient. Communication between the veterinarian, therapist, and owner is important and facilitates the

Table 35-2 Common Medical Conditions Present in Geriatric Dogs, Organized by Organ Systems

Organ System	Condition
Cardiovascular	Cardiomyopathy Endocardiosis Arrhythmias Heartworm disease Pericardial disease Systemic hypertension Endocarditis
Upper respiratory tract	Laryngeal paralysis Brachycephalic respiratory syndrome Laryngeal and pharyngeal neoplasia Bronchomalacia Collapsing trachea
Lower respiratory tract	Pneumonia Neoplasia (primary or metastatic) Pleural effusion Pulmonary edema Bronchitis Pneumothorax
Gastrointestinal	Malabsorptive disease (inflammatory bowel disease) Protein-losing enteropathy Neoplasia (adenocarcinoma, lymphoma)
Hepatobiliary	Hepatic parenchymal disease (chronic hepatitis, cirrhosis, neoplasia, toxic hepatitis) Biliary tract abnormalities (pancreatitis, bile duct neoplasia, cholelithiasis, cholecystitis)
Endocrine	Hyperparathyroidism Hypothyroidism Hyperadrenocorticism Hypoadrenocorticism Thyroid neoplasia Insulinoma Diabetes mellitus Pheochromocytoma Thymoma Hypoparathyroidism
Urinary tract	Renal failure Urinary tract infection Urinary incontinence
Hematologic and immunologic	Anemia (regenerative or nonregenerative) Neoplasia (lymphoma, hemangiosarcoma, leukemia) Hemostatic disorders Immune-mediated polyarthritis Systemic lupus erythematosus
Neurologic	Intervertebral disk disease Cervical spondylomyelopthy Lumbosacral instability Degenerative Myelopathy Neoplasia
Musculoskeletal	Osteoarthritis Neoplasia

assessment of the response to the rehabilitation program and the management of concurrent diseases.

Common Musculoskeletal and Neurologic Considerations

The list of common geriatric diseases is outlined in Box 35-2. The most important systems related to rehabilitation care are the musculoskeletal and nervous systems. Muscle mass and bone mass decrease with aging. Muscle function is diminished as a result of muscle fiber atrophy, loss of elasticity, and reduced oxygen transport to muscles. Cartilage tends to deteriorate as a result of altered biomechanical integrity and loss of tensile strength. The most common problem of the musculoskeletal system in geriatric dogs is OA. Obesity compounds the effects of arthritis (Figure 35-1).[4] Decreased intestinal absorption of calcium and decreased activity level may result in decreased bone mineral content. Osteoporosis, or reduction in the quantity of bone, is rarely clinically significant in aging dogs. The primary importance of senile osteoporosis is its influence on fracture healing. Callus formation is slower in older animals, and fractures require additional healing time. This increases the likelihood of delayed or malunion fractures, muscle atrophy, and disuse osteoporosis. Neoplasia involving the musculoskeletal system, although less common in some breeds of dogs, also must be considered. Primary bone tumors are the most common and include osteosarcoma, chondrosarcoma, fibrosarcoma, and hemangiosarcoma (Figure 35-2).[4]

Several conditions affecting the nervous system also affect aging dogs. In addition to intervertebral disk disease, degenerative conditions such as degenerative myelopathy and spinal neoplasia may occur in older dogs (Figure 35-3). Cognitive dysfunction is a common issue in older dogs. These dogs may exhibit signs such as loss of house training, disturbances in sleep and wake cycles, inattention to food or their environment, and inability to recognize familiar people and places. Short-term memory is affected by changes in interneurons that cause a prolonged effect to a stimulus and increased response time to the external environment.[4] Cognitive dysfunction also may affect the patient's ability to participate in some activities of a rehabilitation program.

Maintaining Quality of Life

A geriatric screening program should be implemented as part of a general wellness program for animals 8 years and older. This allows the veterinarian to target geriatric-related health problems, detect early geriatric disease to implement effective medical care, and offer preventive health care measures. Client education plays a large role in this process. Owners should be instructed on the normal process of aging, nutrition, appropriate exercise, and bereavement counseling, when indicated. A questionnaire is helpful to provide a baseline for function and may be used to reassess the animal's quality of life over time (Figure 35-4).

The normal process of aging includes changes in muscle physiology, hair coat and skin, cognitive function, hearing, and vision. These changes should be assessed, recorded, and addressed as part of the overall treatment program. The initial amount of exercise in the aging dog should be easily tolerated, with increasing amounts added as the rehabilitation program progresses. Low-impact exercise, such as frequent short leash walks and swimming should occur regularly, possibly on a daily basis. Owners should be aware of changes in water or food consumption, rapid weight changes, abnormal urination or defecation, changes in activity level, vomiting, or diarrhea (Figure 35-5). When appropriate, clinicians should advise owners during the final weeks of the animal's life and after their death.[1]

Box 35-2 Common Geriatric Conditions in Dogs

Obesity
Cardiovascular disease
Degenerative joint disease
Cataract(s)
Cancer
Dental disease
Hypothyroidism
Urolithiasis
Hyperadrenocorticism
Diabetes mellitus
Anemia
Urinary incontinence
Hepatopathies
Chronic renal disease

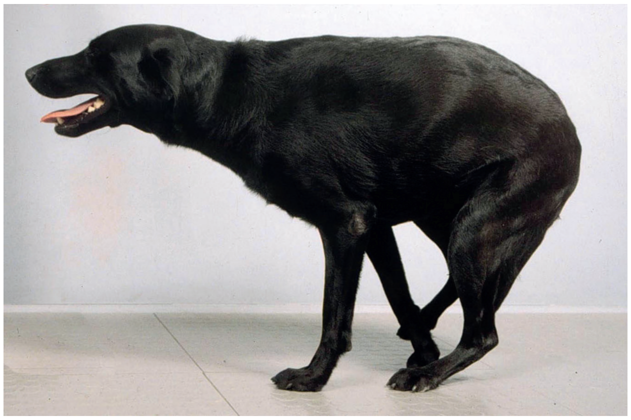

Figure 35-1 The most common problem of the musculoskeletal system in aged dogs is osteoarthritis. Dogs may have difficulty arising, and obesity compounds the effects of arthritis.

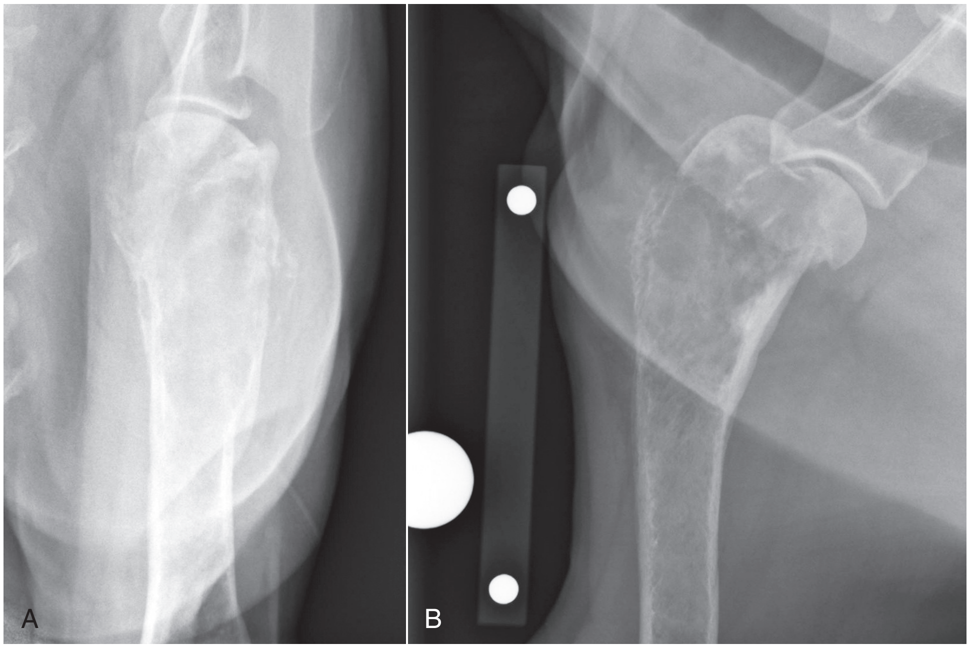

Figure 35-2 This 8-year-old spayed female rottweiler has an osteosarcoma in the proximal portion of the humerus visible on caudocranial **(A)** and mediolateral **(B)** radiographic projections. Classic clinical signs of osteosarcoma include lameness, lethargy, and decreased appetite.

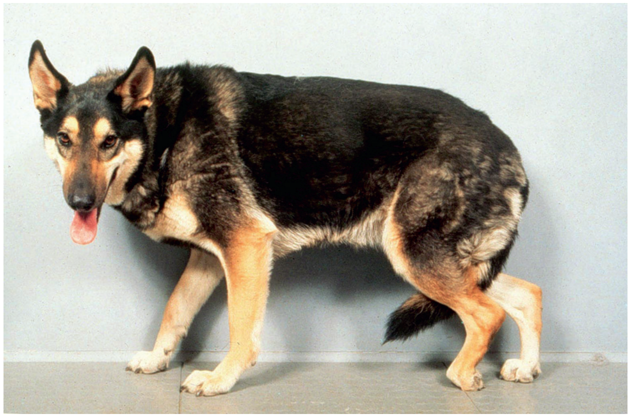

Figure 35-3 Degenerative myelopathy may affect geriatric dogs. The diagnosis is often confounded by the presence of osteoarthritis of multiple joints.

Pain Management in the Geriatric Dog

Because aging dogs may have a fragile physiologic balance, pain management of geriatric dogs should rely as much as possible on therapeutic options with minimal physiologic impact. For example, geriatric people are more sensitive to the gastrointestinal side effects of antiinflammatory medications than younger people.[7] The same most likely applies to dogs.[8] Nonsteroidal antiinflammatory drugs should therefore be used with caution in geriatric dogs. The pain management options with minimal physiologic risks include but are not limited to low-impact exercises, massage, and therapeutic heat and cold. Analgesic medications that may be helpful include amantadine, gabapentin, and tramadol. In patients with severe pain, such as may be present with neoplasia, opioid drugs may be prescribed, or transcutaneous fentanyl patches may be applied for sustained delivery of medication. Although these patches may be helpful in the short term, they are rarely used over extended periods of time because of the cost and other factors involved with prolonged treatment.

Nutritional Considerations

The weight management of geriatric patients is particularly important because obesity is one of the most common conditions in geriatric patients. Because of decreased muscle mass and the possible presence of OA in geriatric dogs, excess weight may predispose aging dogs to critical and potentially irreversible locomotion difficulties.[3] In one lifelong study involving 48 Labrador retrievers, restricted food intake increased median life span and delayed the onset of signs of chronic disease in dogs. In that study littermate dogs were paired, with one dog in each pair fed 25% less food than its littermate, which was fed the same diet ad libitum, from 8 weeks of age until death. Body composition was measured annually until 12 years of age. Compared with control dogs, food-restricted dogs weighed less and had lower body fat content and lower serum triglycerides, triiodothyronine, insulin, and glucose concentrations. Median life span was 18 months longer for dogs in which food was restricted. The onset of clinical signs of chronic disease, including OA, was delayed and generally less severe for food-restricted dogs.

FUNCTIONAL QUESTIONNAIRE

Please complete the questions on this form pertaining to your pet's functional abilities to help us monitor its progress.

1 = not able to perform this activity (needs assistance 100% of the time)
2 = moderate assistance to perform activity (needs assistance >50% of the time)
3 = minimal assistance to perform this activity (needs assistance <50% of the time)
4 = independent with activity (no assistance needed)
5 = N/A

Client:______________________Patient:____________________Date: __________
Medications:____
How often is (are) medication(s) given? ______________________

1. Able to position itself to urinate?	1	2	3	4	5
2. Able to position itself to defecate?	1	2	3	4	5
3. Able to transfer from lying to sitting and vice versa?	1	2	3	4	5
4. Able to transfer from sitting to standing and vice versa?	1	2	3	4	5
5. Able to transfer from lying to standing and vice versa?	1	2	3	4	5
6. Able to roll over?	1	2	3	4	5
7. Able to scratch behind its ears?	1	2	3	4	5
8. Able to ascend stairs?	1	2	3	4	5
9. Able to descend stairs?	1	2	3	4	5
10. Able to walk up an incline/hill?	1	2	3	4	5
11. Able to get in and out of your car?	1	2	3	4	5
12. Able to get on/off a couch or bed?	1	2	3	4	5
13. Able to run?	1	2	3	4	5
14. Able to jump?	1	2	3	4	5

Figure 35-4 Functional questionnaire to assess and monitor the progress of a geriatric patient.

15. Experienced an increase or decrease in weight?	Increase	Decrease	Same
16. Experienced an increase or decrease in endurance?	Increase	Decrease	Same
17. Have you noticed a change in your pet's temperament/attitude? Y N Please elaborate:	Increase	Decrease	Same
18. What does your pet like to do for fun?			
Is he/she able to do that activity? Please elaborate:			
19. Has your pet been able to resume normal activities? Y N Please elaborate:			
20. Able to go on a walk? Y N How long? __________ minutes Could your pet walk longer? Y N Does anything prevent him/her from taking longer walks? Y N If so, what?			
21. Do you notice any problems (limping, stiffness) or are the problems worse after taking a walk? Y N If yes, how so?			
22. Does your pet tire quickly, have to make rest stops, or lag behind during walks? Y N Please elaborate:			

Figure 35-4, cont'd

Continued

23. Does your pet seem to be in pain? Y N

What makes you think this?

24. Is there anything your pet CAN do now that it was NOT able to do before? Y N
Please explain:

LAMENESS SCALE		
0 = Normal stance	0 = No lameness	0 = No lameness
1 = Slightly abnormal stance (partial weight bearing)	1 = Lameness barely perceptible	1 = Lameness barely perceptible
2 = Moderately abnormal stance (toe-touch weight bearing)	2 = Lameness obvious, but not severe	2 = Lameness obvious, but not severe
3 = Severely abnormal stance (holds limb off the floor)	3 = Severe lameness	3 = Severe lameness
4 = Unable to stand	4 = Partial or complete non–weight-bearing lameness	4 = Partial or complete non–weight-bearing lameness

25. Does your dog walk or trot with a limp? (Please select degree of lameness according to the chart below)

Degree of lameness (standing) Degree of lameness (walk) Degree of lameness (trot)

Score: ________________ ________________ ________________

26. Are there any other problems that you have noticed that have not been covered in this form?

27. Additional comments:

Figure 35-4, cont'd

Figure 35-5 Owners should be aware of changes in water or food consumption, rapid weight loss, abnormal urination or defecation, changes in activity level, and vomiting or diarrhea, especially after a major illness or surgery. This dog developed sepsis after an amputation for osteosarcoma.

Figure 35-6 As osteoarthritis progresses, a vicious cycle of pain, reduced activity level, joint stiffness, and loss of strength occurs. The resulting weakness and pain may mimic a neurologic condition.

In addition to restricting calories, adequate protein, minerals, and vitamins also must be present in the diet of the aging patient. Most high-quality commercial diets have adequate nutritional composition to support most phases of life, and some concentrate specifically on aging dogs. Adequate protein is necessary to help maintain muscle mass, and it should be of adequate composition and quality to support muscle maintenance. Concurrent medical conditions, such as heart disease, renal disease, and liver disease, also should receive consideration because dietary management of these conditions is vital to successful management.

Management of the Geriatric Patient with Osteoarthritis

Joint disease is a common problem, affecting up to 20% of adult dogs. Osteoarthritis, often referred to as degenerative joint disease (DJD), is a progressive degenerative condition of joints. Many inflammatory mediators are present in arthritic joints, including interleukins, prostaglandins, and metalloproteinases. There is a progressive cascade of mechanical and biochemical events in patients with OA, which leads to cartilage destruction, subchondral bone sclerosis, synovial membrane inflammation, and periarticular osteophytes. Patients with OA have decreased quality of life, limited activity, reduced performance, muscle atrophy, pain and discomfort, and joint stiffness with decreased range of motion (ROM).[9-11] A vicious cycle of pain, reduced activity level, joint stiffness, and loss of strength occurs as OA progresses (Figure 35-6).

Veterinarians are approached frequently to treat arthritic patients. Traditional management of dogs with OA has included antiinflammatory and analgesic drugs, changes in lifestyle, and surgical management. More recent advances in the management of OA include weight loss, therapeutic exercise, and physical modalities to reduce the severity of clinical signs and reliance on medications to control pain and discomfort.[12,13] Management of the arthritic patient involves a number of modalities and must be tailored to each patient and owner. Nutritional management (including weight control), physical rehabilitation, and medications are the main components for OA management. Cooperation among the veterinarian, therapist, veterinary technician, and owner is vital to carry out an appropriate management program. Regular monitoring of the progress of therapy is essential to help with decision making for further treatment and maintaining enthusiasm for the program. Some of the benefits of a complete program include increasing muscle strength and endurance, increasing joint ROM, decreasing pain, and improving performance, speed, quality of movement, and function. This is particularly important for patients that have become exercise intolerant or have limited mobility.

Much of the pain associated with OA has been attributed to synovitis.[14] The goal of treatment is to provide adequate analgesia to reduce the severity of clinical signs, which allows the patient to be active and undergo therapeutic exercise and weight loss. These treatments are important to accomplish increased muscle strength and joint function, maintain an acceptable quality of life, control pain and discomfort, slow the progression of disease, and promote repair of damaged tissue when possible.

The surgical management of OA focuses on correcting joint disease to prevent further joint degeneration. Early intervention to remove an osteochondritis dissecans lesion, perform a juvenile pubic symphysiodesis or triple pelvic osteotomy, stabilize a cranial cruciate ligament rupture,

Figure 35-7 Early intervention and surgical stabilization of the stifle joint of dogs with a ruptured cranial cruciate ligament may reduce the progression of osteoarthritis.

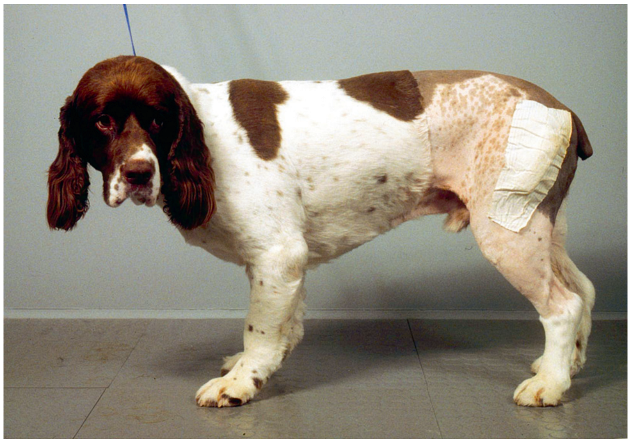

Figure 35-8 Although total hip replacement is a salvage procedure for end-stage osteoarthritis of the hip joint, dogs can have a high quality of life following surgery and rehabilitation.

repair an articular fracture, to reduce hip subluxation associated with hip dysplasia, and correct angular limb deformities to avoid abnormal stresses on joints are all surgical procedures that help slow the degenerative process of affected joints (Figure 35-7). Salvage treatments for severe OA include total joint replacement and arthrodesis of joints, such as the carpus and tarsus (Figure 35-8).

The medical treatment of OA is multifaceted and includes antiinflammatory medications, weight optimization, controlled exercise, physical modalities, alteration of the environment, and disease-modifying OA agents. Veterinarians must impress on owners that the management of chronic OA is a lifelong commitment and is hard work. It is critical to evaluate patients on a regular basis and provide feedback and encouragement to owners. Management of the arthritic patient should be approached in a logical, stepwise progression.

Antiinflammatory Agents

Osteoarthritis is classified as a noninflammatory joint disease because of the relatively low number of inflammatory cells in the synovial fluid. However, many inflammatory mediators (e.g., prostaglandins and leukotrienes), degradative enzymes (metalloproteinases) released as a result of cell membrane injury, and inflammatory cells are intricately involved in a cascade of events leading to the deterioration of articular cartilage. Inflammatory products are produced through enzymatic action (phospholipase A2) on cell membrane phospholipids to produce arachidonic acid. Cyclooxygenase (COX) and lipoxygenase (LOX) act on arachidonic acid to produce prostaglandins, leukotrienes, and related products known as eicosanoids. Nonsteroidal antiinflammatory drugs (NSAIDs) have been the foundation for medical management of OA, particularly in advanced cases. NSAIDs inhibit the COX enzyme, thereby decreasing inflammatory mediators and reducing pain associated with OA. Two isoforms of the COX enzyme have been identified, COX-1 and COX-2. Cyclooxygenase-1 is a constitutive enzyme and is normally produced in physiologic amounts in many tissues. It has protective functions, such as protection of the gastric mucosa, maintaining renal perfusion, and maintaining normal platelet function. The inhibition of the COX-1 enzyme by traditional NSAIDs is believed to be responsible for the adverse side effects of gastric ulceration, prolonged bleeding time, and decreased renal perfusion. The COX-2 enzyme is primarily inducible, and its production increases in response to inflammation. Traditionally NSAIDs have inhibited both COX-1 and COX-2 enzymes. The identification of the two COX isoforms has resulted in research to develop products that selectively inhibit the inducible COX-2 enzyme, while sparing the constitutive COX-1 enzyme. Selective inhibition of COX-2 with preservation of COX-1 could potentially reduce the adverse effects associated with the gastrointestinal tract and kidneys. NSAIDs that are frequently used in veterinary medicine include carprofen, deracoxib, firocoxib, etodolac, and meloxicam. Acetaminophen with or without codeine is occasionally used in dogs, but should never be used in cats. COXIB-class drugs are those that are diarylheterocyclic compounds that preferentially and irreversibly bind the COX-2 enzyme and are relatively COX-1 sparing. Deracoxib and firocoxib are COXIB-class drugs that are approved for use in veterinary medicine. Carprofen, etodolac, and meloxicam also have some preferential selectivity to inhibit mainly COX-2. No NSAID has been shown conclusively to clearly be more efficacious than the others. The effectiveness of various NSAIDs appears to vary between patients.[15] One NSAID could be more effective than another in a patient, and the reverse could be seen for another patient. This means that, for each patient, the optimal NSAID may be identified after several different NSAIDs are used (successively but not concurrently) for 2-week trial periods. A washout period of a few days should be employed, as long as the patient is not in excessive pain during the washout period. It is important to have owners also evaluate the response to treatment because they evaluate the patient in the home environment in which the patient negotiates various obstacles and situations that are not present in a veterinary clinic. Caution must be used when using any NSAID because of their potential side effects. Before prescribing any medication, the patient's physiologic state, especially liver and kidney function, should be assessed. It is important to educate the owners regarding potential side effects, including gastrointestinal events (primarily vomiting and diarrhea), hepatic disease, and renal disease.

Certain forms of polyunsaturated fatty acids (PUFAs), especially omega-3 fatty acids, may reduce the production of certain eicosanoids, especially the more potent inflammatory 4-series leukotrienes, and help reduce inflammation. Some humans with rheumatoid arthritis respond to treatment with PUFAs. Polyunsaturated fatty acids, particularly eicosapentaenoic acid, appeared beneficial to dogs with OA in two studies.[16,17]

Oral or injectable corticosteroids are also prescribed for patients with OA. Steroids are effective antiinflammatory drugs and result in clinical improvement in patients. However, their use as a short-term solution in lieu of other modalities should be avoided because of their probable long-term adverse effects on cartilage and other body systems. For example, glycosaminoglycan (GAG) synthesis by chondrocytes is reduced 30% in patients receiving prednisone. Although veterinarians may feel pressured to provide quick results, client education is critical to emphasize that management of OA is a lifetime endeavor.

Slow-Acting Disease-Modifying Osteoarthritic Agents

Agents thought to alter the course of OA are termed slow-acting disease-modifying osteoarthritic agents because it is thought that they improve the health of the articular cartilage or synovial fluid. These agents may be more beneficial in early OA than in end-stage OA. Nutraceuticals are nutritional supplements believed to have a positive influence on cartilage health by providing precursors necessary for repair and maintenance, and they may also alter the pathophysiology of OA and its progression. Glucosamine and chondroitin sulfate (CS) are routinely combined as disease-modifying agents. Glucosamine is a precursor to the disaccharide units of GAGs, which comprise part of the proteoglycan (PG) articular cartilage matrix. Studies have shown that glucosamine helps to improve cartilage metabolism and upregulates PG synthesis.[18,19] Chondroitin sulfate is the predominant GAG found in articular cartilage. Extracellular and intracellular mechanisms are stimulated by CS to produce GAG and PG. Chondroitin sulfate also competitively inhibits degradative enzymes found in cartilage and synovium.

Polysulfated glycosaminoglycans (PSGAGs) is a drug and is administered as an intramuscular injection. Polysulfated glycosaminoglycans reduce the production of metalloproteinases and increase production of hyaluronic acid and cartilage GAGs. In addition they improve stifle ROM, clinical use of the limb, and the health of the synovium in dogs recovering from experimental cranial cruciate ligament transection and stifle stabilization surgery.[20] Hyaluronic acid (HA) is a nonsulfated GAG and a major component of synovial fluid and cartilage. Hyaluronic acid is available as an intraarticular injection and helps increase synovial fluid viscosity, reduces inflammation and prostaglandin production, and scavenges free radicals. Some patients could potentially benefit from periodic administration of HA.

Obesity

Obesity is strongly associated with the development of OA in people and likely contributes to the progression of OA in dogs.[21,22] For example heavy people are 3.5 times more likely to develop OA than light people, and loss of 5 kg decreases the odds of developing OA by greater than 50%.[23] Additionally, weight loss results in less joint pain and a decreased need for medication to treat OA.[24] In one clinical report weight reduction of 11% to 18% of the initial body weight of obese dogs resulted in improvement of hindlimb lameness associated with hip OA.[25] Another study of obese dogs with hip dysplasia indicated that there is a significant increase in weight bearing as measured with a force plate following weight loss.[26] In addition the owners felt that weight loss resulted in significant improvement in a number of areas in the home environment.

In addition to restricting intake of the normal diet and eliminating treats, prescription diets are available that can help achieve and maintain ideal body weight. In general, a goal is to reduce body fat composition to 20% to 25% of an animal's total body weight. Clinically the ribs should be easily palpable, and there should be a "waist" when the animal is viewed from above and the side.[27]

Physical Rehabilitation Modalities

Mild to moderate exercise and training in normal humans and dogs do not cause OA by themselves, but articular cartilage may undergo biochemical, histologic, and biomechanical changes.[28-31] Most studies of moderate running have indicated that there is no injury to articular cartilage with this form of activity, assuming that there are no abnormal biomechanical stresses acting on the joints. Heavy training programs, however, result in changes that may predispose to the development of OA.[32-34] Controlled therapeutic exercise is a valuable but underused management option for OA patients. Arthritic humans participating in controlled, low-impact exercises have improved function and reduced pain and need for medication.[35,36] The goals of therapeutic exercise should be to reduce body weight, increase joint mobility, and reduce joint pain through the use of low-impact exercises designed to strengthen supporting muscles. Muscle disuse results in atrophy and weakness. Because muscles act as shock absorbers for joints, strengthening periarticular muscles may help protect joints. Mild weight-bearing exercise also helps stimulate cartilage metabolism and increases nutrient diffusion. Exercise may also increase endogenous opiate production; indirectly decreasing OA pain.[37] Exercise programs are tailored to the condition and personality of each patient and to their owners (Box 35-3). An improper program could hasten the progression of OA. Overloading joints should be minimized by performing low-impact activities, such as walking and swimming, until weight loss occurs. Unrealistic demands placed on the owner will likely decrease compliance, and the physical condition of the owner must be considered. Joint instability should be corrected before initiating an exercise program. Exercise programs must be tailored to account for the typical course of exacerbations and remissions of OA. The animal should not be forced to exercise during times of exacerbation of the arthritic condition because inflammation may increase. In preparation for exercising, warming and stretching affected muscle groups and joints during a warm-up period are recommended.[38,39] Tissue warming promotes blood flow to the area, promotes tissue and collagen extensibility, and decreases pain, muscle spasms, and joint stiffness. Heat is contraindicated if swelling or edema is present in a limb or joint. Heating agents such as moist or dry hot packs, circulating warm water blankets, and warm baths typically heat the skin and subcutaneous tissues to a depth of 1 to 2 cm. Another physical agent used for heating is therapeutic ultrasound. Ultrasound frequencies of 1 and 3 MHz in continuous mode produce thermal and nonthermal effects. The effects are related to the treatment time, intensity, frequency, and area being treated. Tissue heating may penetrate to 5 cm, much deeper than with superficial heating modalities. Nonthermal effects include increased cell membrane permeability, calcium transport across the cell membrane, removal of proteins and blood cells from the interstitial spaces, and nutrient exchange. Any stretching should be done during the latter part of warming or immediately after. Massage also has been used to increase blood flow to muscles to warm up the area before activity, and to decrease stiffness after activity. Electrical stimulation of muscle also appears to be of benefit when treating dogs with OA. Transcutaneous electrical stimulation (TENS) was applied to the stifle joint of dogs with experimentally induced OA. Peak vertical force was significantly increased after TENS application in these dogs.[40]

Controlled leash walking, walking on a treadmill, jogging, swimming, and going up and down stairs or ramp are excellent low-impact exercises (Figure 35-9). The length of the exercise should be titrated so there is no increased pain after activity. Also exercise sessions should be short in the early phases of training: three 10-minute sessions is preferred to one 30-minute session. Walks should be brisk and purposeful, minimizing stopping. Avoiding sudden bursts of activity will help avoid acute inflammation of arthritic joints. Swimming and walking in

Box 35-3 Guidelines for an Osteoarthritis Home Care Program

Osteoarthritis tends to follow a course of exacerbation and remission. During a period of aggravation, do not force exercise. Low-impact exercises, such as leash walking and swimming, are ideal.

One sign of osteoarthritis is stiffness, especially in the morning. Begin the day by taking time to warm up the joints. This can be done using warm water bottles, hot packs, heating wet towels in the microwave (comfortable enough for you to tolerate on your own skin), or using a bathtub/whirlpool (≈85°) for 15 to 20 minutes.

Following warm-up, apply passive range of motion to the affected joints. Slowly flex and extend the joints—DO NOT force beyond a comfortable range. Fifteen to twenty SLOW repetitions are sufficient. This helps to decrease stiffness and increase motion in the joints.

Continue throughout the day taking the dog on several short walks (at least three) with long rest periods in between. Let the patient set the time and distance. Begin walking 10 to 15 minutes at a time, gradually increasing the time and length. Multiple short walks are more beneficial than one or two long walks. Note any discomfort, increased fatigue or stiffness, or refusal to go any further. Should this occur, cut the previous level of activity in half and continue from there.

Differentiate between muscle soreness and joint pain:

- Muscle soreness—painful upon palpation of muscle groups
- Joint pain—painful when flexing or extending a joint

As tolerance and endurance improve, activities to further improve mobility, reduce pain, and increase muscle strength may be added. Some ideas may be to incorporate ramps, inclines, declines, or stairs into the walking program in incremental amounts. Once again, let the patient set the boundaries.

Following exercise, a period of cool-down is necessary. Massage and use a bag of frozen vegetables or an ice pack on the joints for 15 to 20 minutes to minimize any postexercise swelling that may occur.

No antiinflammatory medications should be given for the first day when beginning a rehabilitation program. The antiinflammatory effect may mask painful joints associated with exercise progression. After a day if no joint pain is noticed, medications may be given 1 hour before exercise. Withhold medication again as exercise is increased to be certain that the medication is not masking excessive pain.

Osteoarthritis treatment is a lifelong commitment. If rehabilitation is discontinued, any benefits gained will likely be lost.

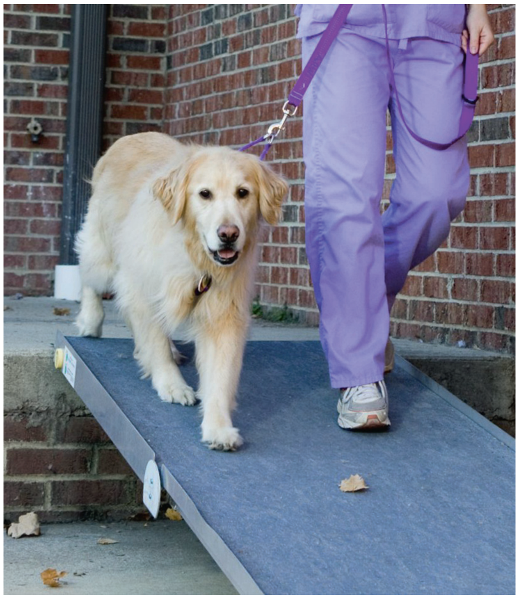

Figure 35-9 Going up and down an inclined ramp is an excellent low-impact exercise. A handicap access ramp also makes it easier to get into buildings.

Figure 35-10 Training on an underwater treadmill may significantly increase weight bearing comparable to achievements obtained using medication in many patients.

water (Figure 35-10) are some of the best activities for dogs. The buoyancy of water is significant and limits the impact on the joints while promoting muscle strength and joint motion. If joint pain is perceived to be greater after exercising, the length of the activity can be decreased by half. When stepping up the amount of activity, the increase should be approximately 10% to 15%, once a week unless increased pain or decreased function occurs. Ideally, anti-inflammatory drug use should be minimal and consideration should be given to avoid administration immediately before stepping up activity because it is important to determine if the level of exercise is too great and causes pain. The exercise periods should be evenly spaced throughout each day and over the entire week. Training and conditioning help maintain an ideal body weight, improve ROM, and increase muscle strength and tone, which help to stabilize joints. Following exercise a 5 to 10-minute cool-down period is recommended. A slower paced walk may be initiated for 3 to 5 minutes, followed by ROM and stretching exercises. A cool-down massage may help decrease pain, swelling, and muscle spasms. Finally, cryotherapy (cold packs or ice wrapped in a towel) may be applied to painful areas for 15 to 20 minutes to control postexercise inflammation. Application of cold decreases blood flow, inflammation, hemorrhage, and metabolic rate.

Environmental Modifications

Altering the environment may be helpful for dogs with moderate to severe arthritis. The principles of environmental modifications for dogs are similar to those for arthritic humans.[41] Whenever possible animals should be moved from a cold, damp environment (i.e., outdoor) to a warm, dry environment (indoor). A soft, padded bed should be provided. A circulating warm-water blanket under the blankets provides heat that may reduce morning stiffness. Provide good footing to avoid slipping and falling. Minimize stair climbing through the use of handicapped ramps and keeping pets on ground floors. Steps are negotiated more easily if they are wider, have a gradual rise, and are spaced farther apart. Portable ramps are available to assist patients getting in and out of vehicles. Avoid overdoing activities on the weekends, and prevent excessive play with other pets because arthritic animals may attempt to keep up, and in the process, become more lame and painful. In some instances, however, play with other animals stimulates activity and provides a welcome break in the exercise routine.

Rehabilitation for the Patient with Neoplasia

Geriatric patients with neoplasia may benefit from specific rehabilitation programs. For example fibrosis of the skin and subcutaneous tissue may result from radiation therapy. When such fibrosis involves limb tissues, the ROM of a joint may be limited and limb use may be negatively impacted. A gentle massage of the affected region, stretching, and ROM exercises may help limit the impact of fibrosis. It is particularly critical to develop a comprehensive pain management program in cancer patients because the chronic pain associated with some neoplasms negatively affects the quality of life of patients and may lead to anorexia and decreased ambulation. In addition to all the conventional strategies used for pain management, fentanyl patches may be used to provide pain relief in the short term. Palliative radiation therapy may also be used to provide pain relief in a limb with osteosarcoma. Although